

Varun Reddy Gopavaram

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Tampa, FL (Open to Relocate)

Professional Summary:

Results-driven **Data Analyst** with **5+** years of experience delivering end-to-end data solutions across healthcare, e-commerce, and tech sectors. Proven track record in building AI agents, designing **ETL** pipelines, developing predictive models, and creating dynamic dashboards using tools like **Python**, **SQL**, **Power BI**, and **AWS**. Specializing in developing AI agents, deploying **LLMs (GPT-4, LLaMA)**, and optimizing **NLP** pipelines using frameworks like **LangChain** and **FAISS**. Skilled in transforming complex data into actionable insights to drive strategic decision-making and operational efficiency. Adept at driving data-driven decision-making and improving operational efficiency. **Actively seeking opportunities** to contribute to impactful, data-driven business solutions.

Technical Skills:

- **Databases:** Snowflake, Microsoft SQL Server, MySQL
- **Programming Languages:** Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch), HTML, CSS, JavaScript, R, Power Query, DAX
- **Cloud:** AWS (S3, Glue, Lambda, Redshift, Athena), IAM, VPC, Azure
- **Tools & Libraries:** Excel, PowerPoint, JIRA
- **Data Visualization:** PowerBI, Tableau, Tableau Prep Builder, Alteryx
- **Machine Learning:** Supervised ML (Linear and Logistic Regression, Decision Trees, Xgboost), Unsupervised ML (K-means), NLP, Classification techniques (binary classification, multi-class classification), A/B testing
- **ML/NLP:** OpenAI, LangChain, RAG, BERT, LLaMA, FAISS, Pinecone, PhiData, MCP, Hugging Face

Professional Experience:

RIGHTPATH RESEARCH AND INNOVATION CENTER, USF

Tampa, FL

Data Analyst

Nov 2023 - May 2025

Worked with researchers studying child behavior using platforms like REDCap and Inquisit (Azure Blob Storage) to collect data across 50+ studies. Focused on the technical side of the research team — cleaning and analyzing data, and building predictive models to help researchers draw meaningful conclusions. In the last six months, we improved research efficiency by developing AI agents that enabled natural language interaction with data.

- Built a research-facing AI agent integrated with **Pinecone** and **GPT-4**, enabling researchers to semantically search behavioral health documents via natural language prompts, reducing manual data lookup time by **60%**
- Utilized advanced **SQL** skills to write complex queries, and construct procedures, functions, and workflows that supported application development, reporting, and data extraction processes
- Leveraged **Python** to meticulously clean, pre-process, and manipulate datasets, ensuring a high standard of data quality and preparing them appropriately for thorough analysis
- Processed and embedded over 10k pages of behavioral study documents using **MiniLM** embeddings and stored them in **Pinecone's** vector index, accelerating semantic retrieval across 10+ research projects
- Deployed **RAG** workflow powered by **LangChain**, **Pinecone**, and **OpenAI**, leveraging advanced prompt engineering techniques to generate context-aware answers—boosting researcher satisfaction by **30%**
- Centralized behavioral research data by building SSIS-based **ETL pipelines** to ingest from **Azure Blob Storage** into **SQL Server**, enhancing data access and usability for 50+ ongoing research projects
- Developed a sentiment analysis UDF in **Snowflake** to detect positive, negative, or neutral sentiment in open-ended survey responses, enabling researchers to extract insights from qualitative data

- Applied feature engineering on 25+ behavioral indicators (e.g., attention span, emotional regulation), increasing model interpretability for early childhood intervention analysis
- Trained and tuned **Logistic Regression** and **XGBoost** models on 30K+ assessments using Stratified K-Fold, boosting F1 score by 22% and improving precision/recall in predicting post-intervention outcomes
- Designed interactive **Power BI** dashboards to monitor 15+ behavioral KPIs, reducing manual reporting time by 20% and enabling data-driven insights for researchers

AMAZON

Hyderabad, India

Data Analyst

Nov 2021 - Jun 2023

Amazon was losing time and money—millions in dwell fees—because freight loads were getting stuck between rail yards in the Midwest. I was brought in to analyze shipment data and reduce dwell charges and turnaround time. With the new system I helped build, we cut turnaround time from over 96 hours to under 24 hours and slashed dwell fees from \$1M per year to under \$ 20 K.

- Put together a step-by-step plan using Excel, internal wiki tools, and Amazon Chime to organize tasks, align teams, and prioritize key data, helping the team move faster and roll out solutions smoothly
- Built serverless data pipelines using AWS Lambda and Data Wrangler to convert JSON to Parquet, catalog data in Glue, and enable real-time S3 ingestion, reducing refresh time by 90%
- Performed EDA on routing strategies using SQL in **AWS** Athena, uncovering bottlenecks in carrier performance and reducing shipment delays by 45%
- Conducted **A/B testing** on routing strategies using **SQL** and **Python** to evaluate impact on delay rates, uncovering a new rule that reduced shipment exceptions by 20%.
- Validated and reconciled data across carrier and logistics systems, enhancing data accuracy and boosting the reliability of enterprise reporting.
- Designed Excel-based planning tools with **XLOOKUP**, **PivotTables**, and **Macros** to automate carrier emails and generate performance reports, reducing daily planning time from 6+ hours to 2 hours
- Automated **Power BI** and **QuickSight** dashboards to track logistics KPIs like on-time delivery and exceptions, cutting manual reporting by 75% and improving daily stakeholder visibility

BRIGHTCHAMPS TECH

Hyderabad, India

Data Analyst

Jan 2021 - Oct 2021

Worked on extracting and transforming data from ad campaigns across multiple channels such as Instagram, Facebook, Google Analytics, and email. This data was used to support smarter, data-driven decisions about marketing budget allocation. Also focused on reducing user churn by analyzing sign-up sources and predicting which users were likely to drop off, enabling the team to take early action to retain them

- Engineered data pipelines using **SQL** (MySQL) to extract and analyze user journey metrics—including visits, enrollments, and payments—resulting in actionable insights that shaped marketing and growth strategies.
- Built **Power BI** dashboards to track **KPIs** such as CAC, LTV, funnel conversion rates, and regional performance; reduced reporting turnaround by 50%
- Conducted funnel and churn analysis; identified high-risk behaviors and enabled targeted interventions, resulting in a 12% reduction in early-stage churn
- Mapped the complete sales funnel by integrating data from Google Analytics, Facebook Ads, and HubSpot CRM; identified a 15% drop-off at the demo stage and optimized follow-up strategy to improve conversions
- Designed and analyzed **A/B tests** on landing pages and email subject lines using Google Optimize and HubSpot; surfaced high-performing variants that boosted click-through rates by 18% and increased lead-to-signup conversions
- Delivered weekly insights to marketing and product leadership, influencing decisions on campaign spend, onboarding flows, and engagement tactics

- Partnered with developers to enhance custom event tracking via **Google Tag Manager**; ensured 100% accuracy in key funnel events, improving attribution reliability and data confidence
- Collaborated with developers, marketers, and product managers to enhance data quality, optimize tracking logic, and increase system usability across platforms

INTELUX ELECTRONICS PVT. LTD

Pune, India

IT Intern

Jan 2018 - Jun 2018

Intelux Electronics, a maker of energy-efficient products, was developing a web app to enhance its digital presence. Supported the project by helping with front-end design and backend development using HTML, CSS, JavaScript, PHP, and MySQL

- Assisted in developing Intelux's web app using **HTML, CSS, JS, and PHP** to improve the company's digital presence
- Worked with WordPress and third-party plugins (e.g., RevSlider) to configure homepage sliders, product showcases, and interactive UI components under the guidance of the senior web developer
- Implemented responsive layouts with **Bootstrap**, enhancing mobile usability and overall user experience
- Built backend modules in **PHP** and **MySQL** to manage product listings and automate contact form handling

Projects:

AI Chatbot | NLP, LLM

- Launched an AI-powered equity research chatbot using LangChain and OpenAI, cutting manual analysis time by 60%
- Optimized **RAG** with **FAISS** and **OpenAI** Embeddings, enabling 70% faster search while cutting API costs by 50%

Analysis of Advanced Database Management System for E-commerce

- Architected a scalable e-commerce database schema using advanced SQL data modeling (normalization, partitioning, indexing), ensuring ACID compliance and handling high-volume transactional workloads.
- Implemented comprehensive **ETL** processes to clean, transform, and migrate product and customer data from legacy systems to cloud infrastructure.
- Automated stock tracking and order management processes through stored procedures and triggers, integrating with external API endpoints to synchronize with inventory management and fulfillment

AWARDS, RECOGNITION, CERTIFICATIONS:

- [Power BI Data Analyst Associate](#)
- Amazon Rockstar award Q3 2022, Q1 2023, Amazon Spot award Q1 2023
- Successfully completed HIPAA Privacy Training at USF Health (May 2024, May 2025) to support compliance in healthcare research and data handling environments

Education:

UNIVERSITY OF SOUTHERN FLORIDA

Tampa, FL

MS in Business Analytics and Information Systems (GPA:3.93/4)

Aug 2023 - May 2025

ICFAI Business School (IBS)

Hyderabad, India

MBA

Jun 2019 - Mar 2021

Symbiosis International University

Pune, India

Bachelor of Technology in Information Technology

Jul 2015 - Mar 2019